



Bahir Dar University
Bahir Dar Institute of Technology



Engineering Thermodynamics I

Chapter 3: Work and Heat

Outlines

- ❖ Work; units of work
- ❖ Expansion and compression work
- ❖ Other forms of work
- ❖ Heat; units of work

Forms of Energy

- Energy can exist in numerous forms such as thermal, mechanical, kinetic, potential, electric, magnetic, chemical, and nuclear, and their sum constitutes the total energy E of a system.

$$e = \frac{E}{m} \quad (\text{kJ/kg})$$

- Thermodynamics provides no information about the absolute value of the total energy. It deals only with the change of the total energy. Various forms of energy that make up the total energy of a system in two groups: **macroscopic and microscopic**. **Macroscopic forms** of energy are those a system possesses as a whole **with respect to some outside reference frame**, such as kinetic and potential energies. The **microscopic forms** of energy are those **related to the molecular structure of a system and the degree of the molecular activity**, and they are independent of outside reference. Internal energy of a system is sum of all the microscopic forms of energy.
- From macroscopic forms:

$$KE = m \frac{V^2}{2} \quad (\text{kJ})$$

$$ke = \frac{V^2}{2} \quad (\text{kJ/kg})$$

$$PE = mgz \quad (\text{kJ})$$

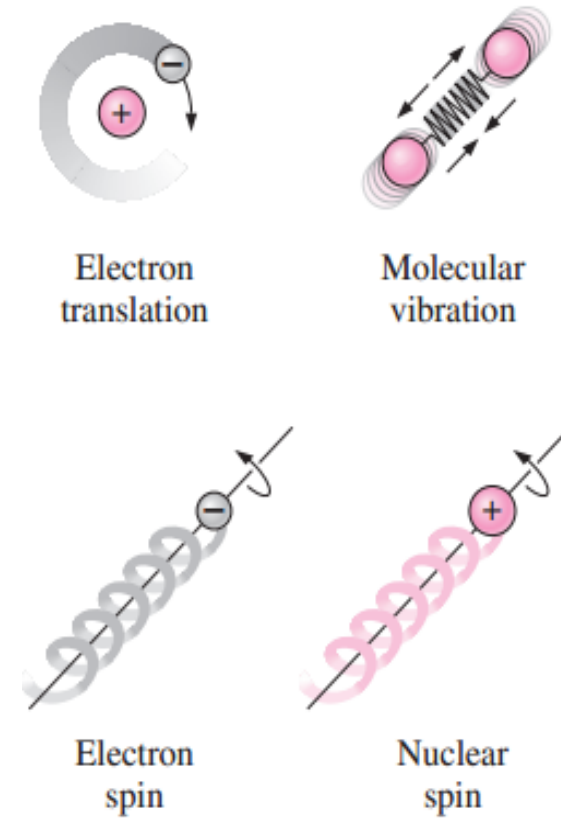
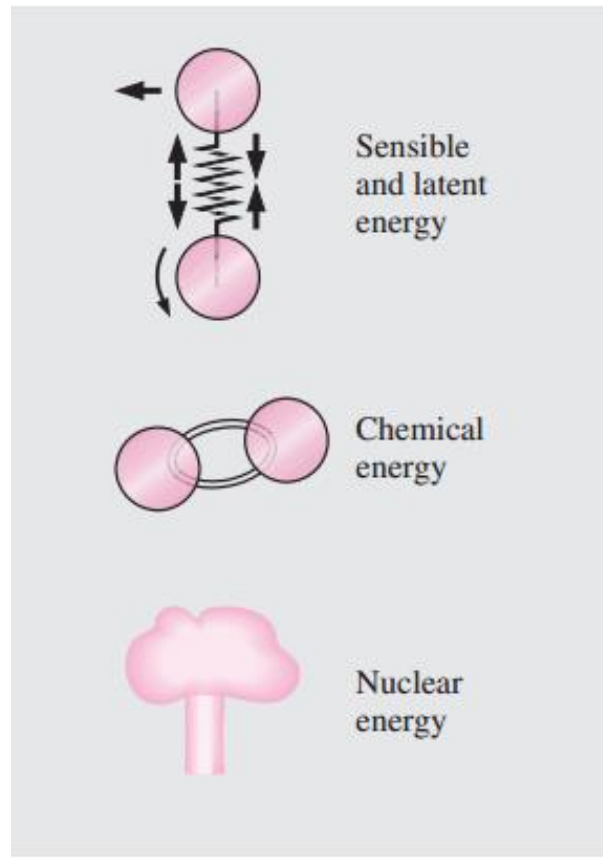
$$pe = gz \quad (\text{kJ/kg})$$

- In the absence of magnetic, electric, and surface tension effects, the total energy of a system consists of the kinetic, potential, and internal energies and is expressed as:

$$E = U + KE + PE = U + m \frac{V^2}{2} + mgz \quad (\text{kJ}) \qquad e = u + ke + pe = u + \frac{V^2}{2} + gz \quad (\text{kJ/kg})$$

Some Physical Insight to Internal Energy

- The change in the total energy ΔE of a stationary system is identical to the change in its internal energy ΔU . The portion of the internal energy of a system associated with the **kinetic energies of the molecules** is called the **sensible energy**. Kinetic energy due to translational energy, rotational energy, vibrational energy, and spin energy of molecules and/or electrons/particles of nucleus. The **average velocity** and the **degree of activity** of the molecules are **proportional to the temperature of the gas** (higher temperature means the molecules possess higher kinetic energies, higher U).
- The internal energy is also associated with various binding forces. The internal energy associated with the phase of a system is called the **latent energy**.
- U also include chemical energy and nuclear energy.



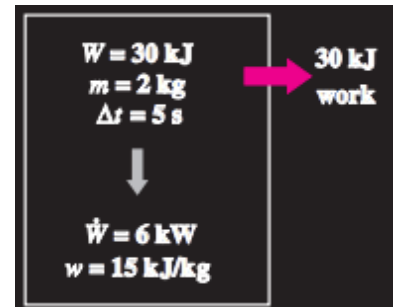
- The forms of **energy not stored** in a system can be viewed as the **dynamic forms of energy** or as energy interactions. The dynamic forms of energy are **recognized at the system boundary as they cross it**, and they represent the **energy gained or lost** by a system during a process. The only two forms of energy interactions associated with a **closed system** are **heat transfer and work**.

Energy Transfer by Work and heat

- Energy can cross the boundary of a closed system in the form of **work** and **heat**
- If the energy crossing the boundary of a closed system is not heat, it must be work since, heat is easy to recognize: Its driving force is a temperature difference. More specifically, **work** is the energy transfer associated with a force acting through a distance.

$$W = \int_1^2 F dx$$

- From thermodynamics point of view, work is done by the system if the sole effect on the surrounding could be raising of a weight.
- Work is an energy and its units is kJ

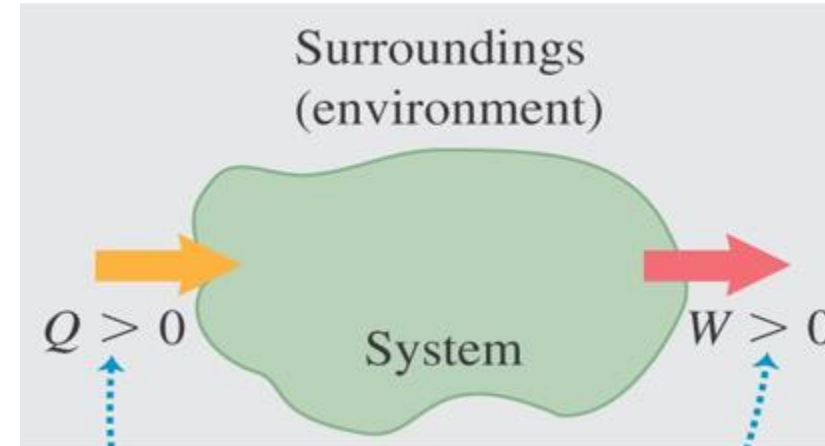
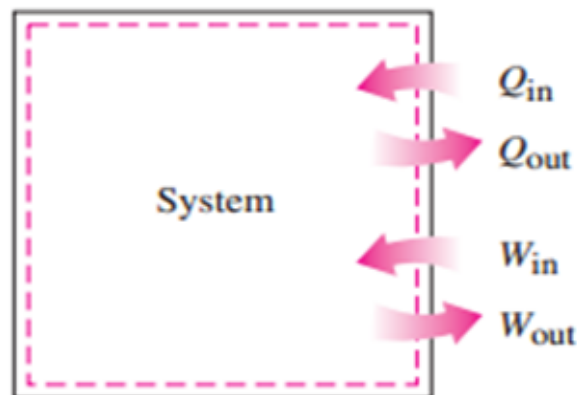


specific work and time rate of work(P)

- Heat and work are directional quantities; heat transfer to a system and work done by a system are positive; heat transfer from a system and work done on a system are negative.

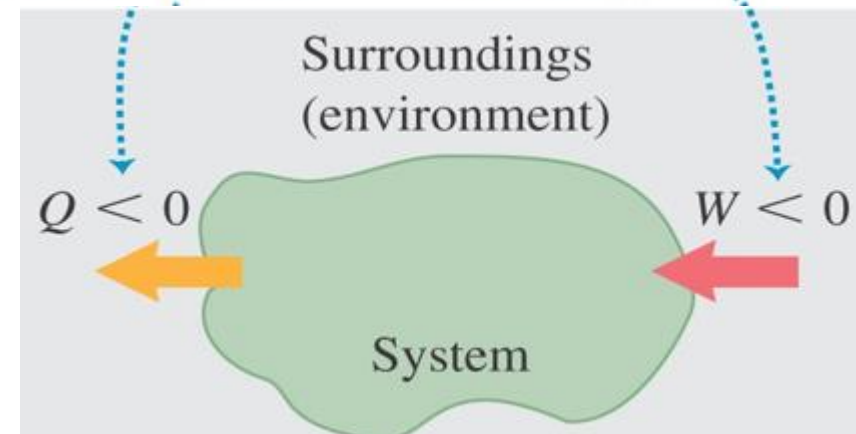
Sign convention

- ❖ Heat transfer to the system-
 Q_{in} is **positive**
- ❖ Heat transfer from the system-
 Q_{out} is **negative**
- ❖ W done by the system is **positive**
- ❖ W done on the system is **negative**



Heat is positive when it *enters* the system, negative when it *leaves* the system.

Work is positive when it is done *by* the system, negative when it is done *on* the system.



Similarities between Work and heat:

- Both are recognized at the boundaries of a system as they cross the boundaries. That is, both heat and work are **boundary phenomena**.
- Systems possess energy, but not heat or work.
- Both are associated with a process, not a state. Unlike properties, heat or work has no meaning at a state.
- Both are path functions (i.e., their magnitudes depend on the path followed during a process as well as the end states).

- Work is a path function and inexact differential.

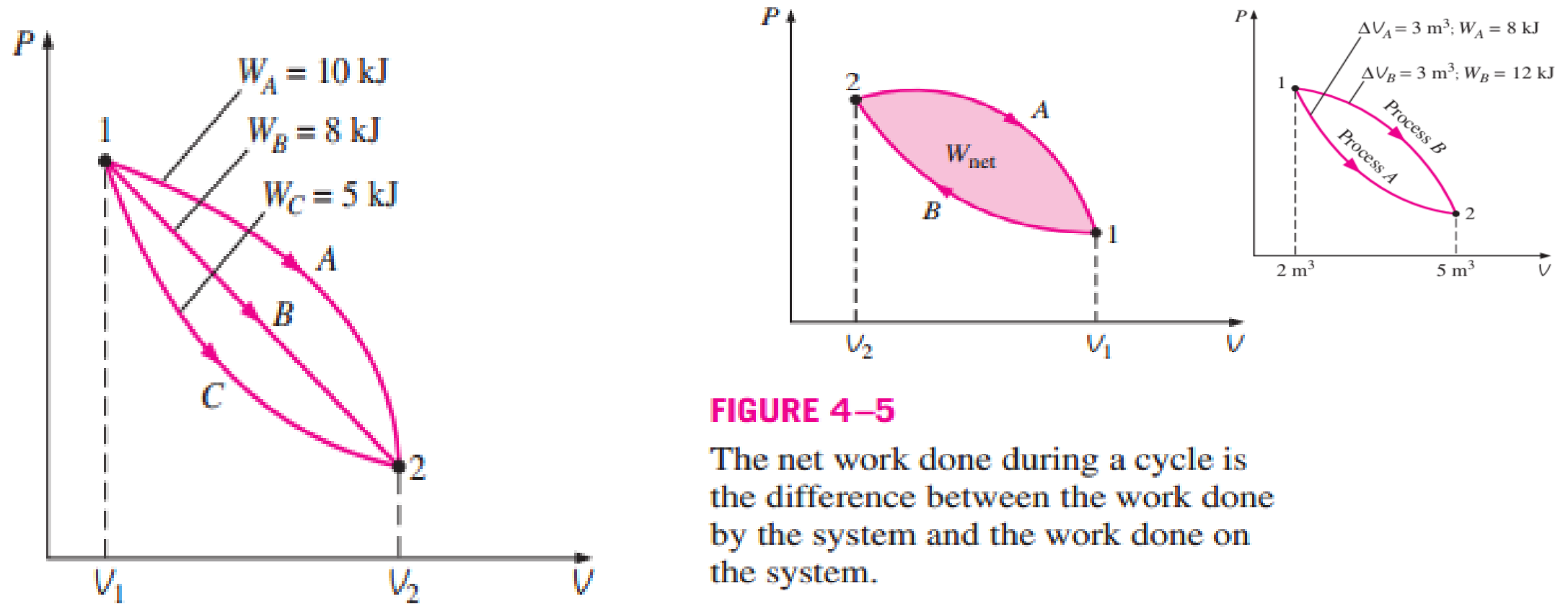


FIGURE 4-5

The net work done during a cycle is the difference between the work done by the system and the work done on the system.

$$\int_1^2 \delta W = W_{12} \quad (\text{not } \Delta W) \quad \text{Inexact differential}$$

$$W_b = W_{\text{friction}} + W_{\text{atm}} + W_{\text{crank}} = \int_1^2 (F_{\text{friction}} + P_{\text{atm}}A + F_{\text{crank}}) dx$$

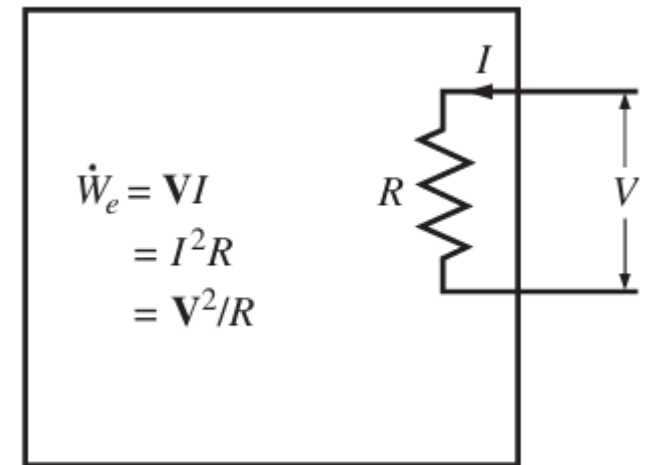
Forms of work:

Electrical Work:

- In an electric field, electrons in a wire move under the effect of electromotive forces, doing work.
- The electrical work done is $W_e = \mathbf{V}N$
- In general, both V and I vary with time, and the electrical work done during a time interval

$$W_e = \int_1^2 \mathbf{V}I dt \quad (\text{kJ})$$

- When both V and I remain constant during the time interval t , it reduces to $W_e = \mathbf{V}I \Delta t \quad (\text{kJ})$



Forms of work:

Mechanical forms of work: many thermodynamic problems involve this kind of work. The work done by a constant force F on a body displaced a distance s in the direction of the force is given by:

$$W = Fs \quad (\text{kJ})$$

$$W = \int_1^2 F ds \quad \text{If } F \text{ is not constant}$$

There are two requirements for a work interaction between a system and its surroundings to exist:

(1) there must be a force acting on the boundary, and (2) the boundary must move.

It is associated with the movement boundary or movement of a system as a whole (must be force acting on the boundary or the boundary must move).

Some common forms of mechanical work are:

Shaft Work: Energy transmission with a rotating shaft

$$T = Fr \rightarrow F = \frac{T}{r}$$

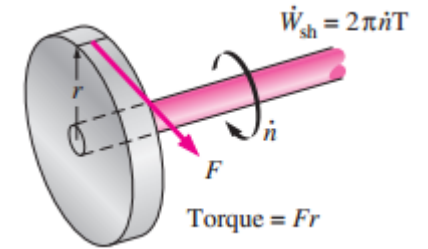
This force acts through a distance s , which is related to the radius r by

$$s = (2\pi r)n$$

Then the shaft work is determined from

$$W_{\text{sh}} = Fs = \left(\frac{T}{r}\right)(2\pi rn) = 2\pi nT \quad (\text{kJ})$$

$$\dot{W}_{\text{sh}} = 2\pi \dot{n}T \quad (\text{kW})$$



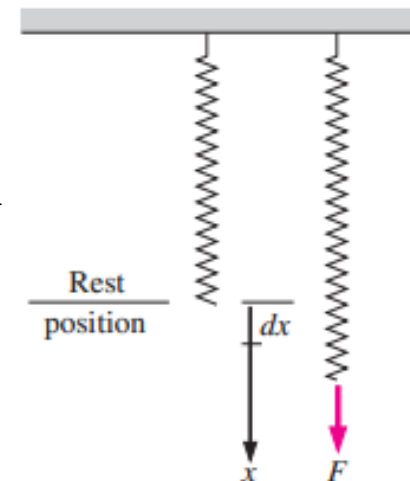
Spring work: the length of the spring changes by a differential amount dx under the influence of a force F , the work done is

$$\delta W_{\text{spring}} = F dx$$

Linear elastic springs, the displacement x is proportional to the force applied

$$F = kx$$

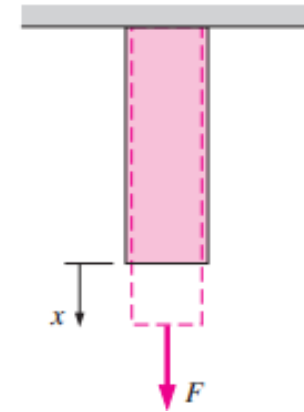
$$W_{\text{spring}} = \frac{1}{2}k(x_2^2 - x_1^2)$$



Work Done on Elastic Solid Bars: Solids are often modeled as linear springs because under the action of a force they contract or elongate, and when the force is lifted, they return to their original lengths, like a spring.

$$\text{normal stress } \sigma_n = F/A,$$

$$W_{\text{elastic}} = \int_1^2 F \, dx = \int_1^2 \sigma_n A \, dx$$



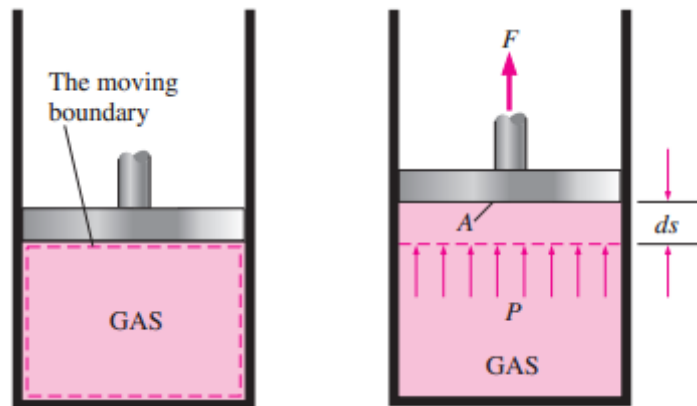
Work Done to Raise or to Accelerate a Body: When a body is raised in a gravitational field, its potential energy increases. Likewise, when a body is accelerated, its kinetic energy increases. The conservation of energy principle requires that an equivalent amount of energy must be transferred to the body being raised or accelerated. The work transfer needed to raise a body is equal to the change in the potential energy of the body, and the work transfer needed to accelerate a body is equal to the change in the kinetic energy of the body.

Non mechanical Forms of Work: non mechanical work modes can be treated in a similar manner by identifying a generalized force F acting in the direction of a generalized displacement x . Some examples of non mechanical work modes are electrical work, magnetic work, electrical polarization work.

- Electrical work: Force $\Rightarrow$ voltage, displacement $\Rightarrow$ electrical charge
- Magnetic work: Force $\Rightarrow$ magnetic field strength, displacement $\Rightarrow$ magnetic dipole moment
- Electrical polarization work: Force $\Rightarrow$ electric field strength, displacement $\Rightarrow$ polarization of the medium.

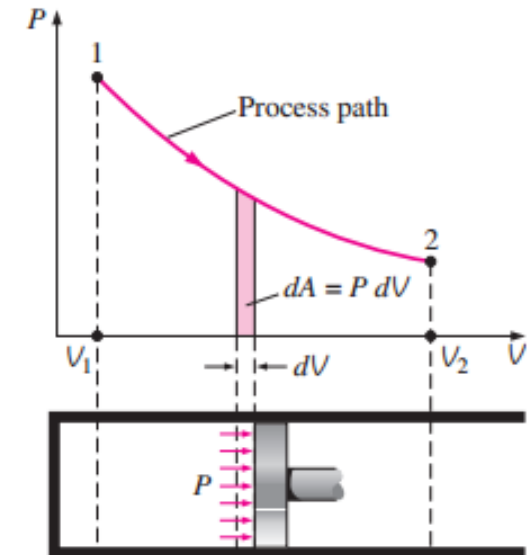
Moving Boundary Work

- One form of mechanical work frequently encountered in practice is associated with the expansion or compression of a gas in a piston–cylinder device. The expansion and compression work is often called moving boundary work, or simply boundary work which mostly involved in automobile engines. Work, being a **path function**, cannot be determined analytically without a knowledge of the path for real engines due to fast compression. Therefore, the boundary work in real engines or compressors is determined by direct measurements.



$$\delta W_b = F ds = PA ds = P dV$$

$$W_b = \int_1^2 P dV \quad (\text{kJ})$$



$$\text{Area} = A = \int_1^2 dA = \int_1^2 P dV$$

Moving boundary work for a quasi-equilibrium process

Boundary work also known as Pdv work.

$$W_b = W_{\text{friction}} + W_{\text{atm}} + W_{\text{crank}} = \int_1^2 (F_{\text{friction}} + P_{\text{atm}}A + F_{\text{crank}}) dx$$

Polytropic Process

- During actual expansion and compression processes of gases, pressure and volume are often related by $PV^n = C$ where n and C are constants. A process of this kind is called a Polytropic process. Expressing P in terms of V :

$$P = CV^{-n}$$

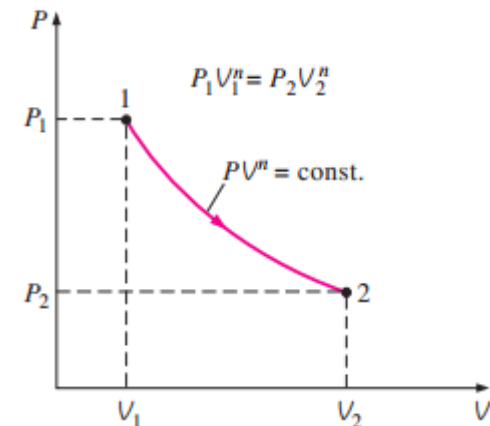
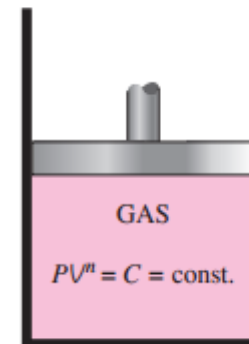
$$W_b = \int_1^2 P dV = \int_1^2 CV^{-n} dV = C \frac{V_2^{-n+1} - V_1^{-n+1}}{-n+1} = \frac{P_2 V_2 - P_1 V_1}{1-n}$$

since $C = P_1 V_1^n = P_2 V_2^n$. For an ideal gas ($PV = mRT$), this equation can also be written as

$$W_b = \frac{mR(T_2 - T_1)}{1-n} \quad n \neq 1 \quad (\text{kJ})$$

For the special case of $n = 1$ the boundary work becomes

$$W_b = \int_1^2 P dV = \int_1^2 CV^{-1} dV = PV \ln\left(\frac{V_2}{V_1}\right)$$



Energy transfer by Heat

Heat is defined as the form of energy that is transferred between two systems (or a system and its surroundings) by virtue of a temperature difference. A process during which there is no heat transfer is called an adiabatic process. Heat is also a path function so that it doesn't depend on the initial and final state of the system. It is also an inexact differential.

$$Q = \int_{t_1}^{t_2} \dot{Q} dt$$

Heat is transferred by three mechanisms: conduction, convection, and radiation.

- **Conduction:** is the transfer of energy from the more energetic particles of a substance to the adjacent less energetic ones as a result of interaction between particles.
- **Convection:** is the transfer of energy between a solid surface and the adjacent fluid that is in motion, and it involves the combined effects of conduction and fluid motion.
- **Radiation:** is the transfer of energy due to the emission of electromagnetic waves (photons).

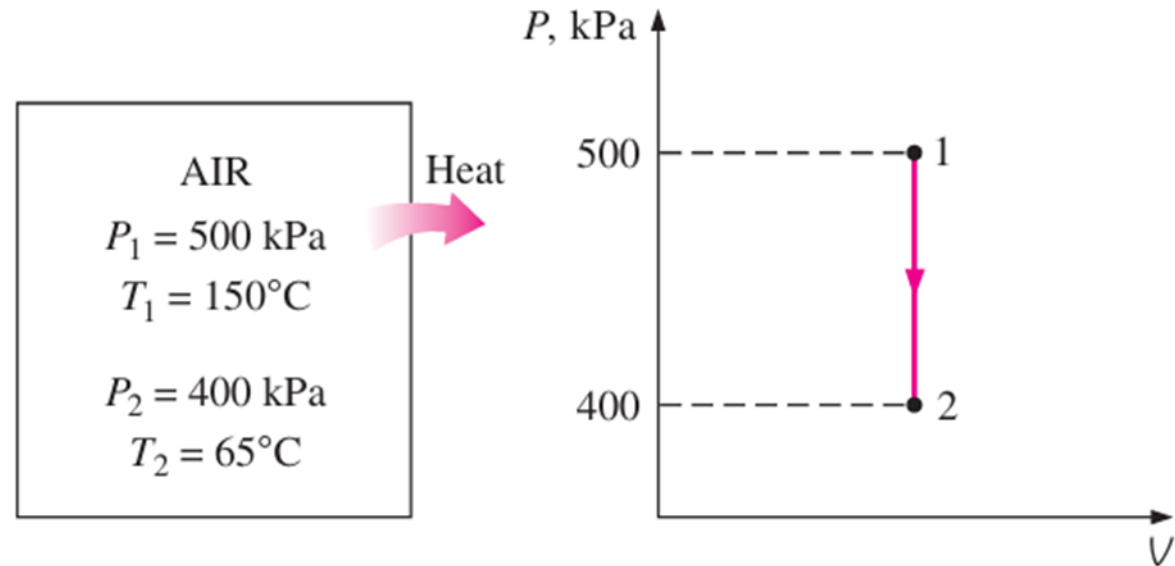
Example:3.1

A rigid tank contains air at 500 kPa and 150°C. As a result of heat transfer to the surroundings, the temperature and pressure inside the tank drop to 65°C and 400 kPa, respectively. Determine the boundary work done during this process.

Solution

Air in a rigid tank is cooled, and both the pressure and temperature drop. The boundary work done is to be determined. Analysis A sketch of the system and the P-V diagram of the process are shown in Fig. The boundary work can be determined from

$$W_b = \int_1^2 P dV \overset{0}{=} 0$$



Discussion This is expected since a rigid tank has a constant volume and $dV=0$ in this equation. Therefore, there is no boundary work done during this process.

Example:3.2

A piston–cylinder device initially contains 0.4 m^3 of air at 100 kPa and 80°C . The air is now compressed to 0.1 m^3 in such a way that the temperature inside the cylinder remains constant. Determine the work done during this process.

Solution Air in a piston–cylinder device is compressed isothermally. The boundary work done is to be determined.

Analysis A sketch of the system and the P - V diagram of the process are shown

Assumptions 1 The compression process is quasi-equilibrium. 2 At specified conditions, air can be considered to be an ideal gas since it is at a high temperature and low pressure relative to its critical-point values.

Analysis For an ideal gas at constant temperature T_0 ,

$$PV = mRT_0 = C \quad \text{or} \quad P = \frac{C}{V}$$

where C is a constant. Substituting this into

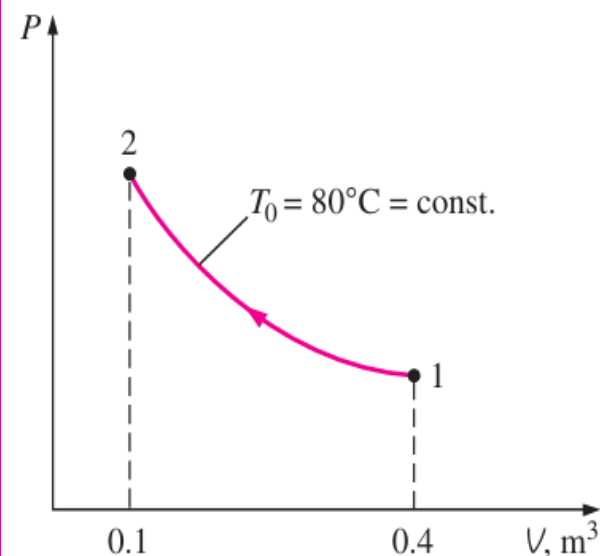
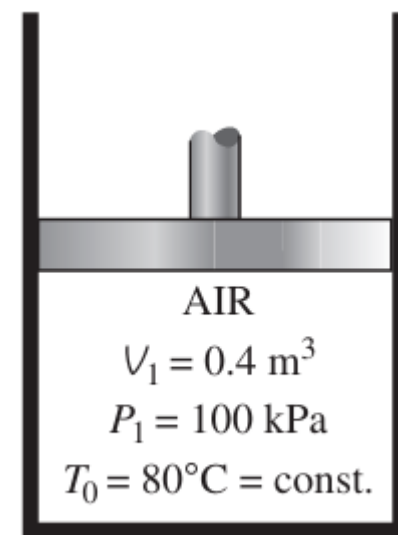
$$W_b = \int_1^2 P dV = \int_1^2 \frac{C}{V} dV = C \int_1^2 \frac{dV}{V} = C \ln \frac{V_2}{V_1} = P_1 V_1 \ln \frac{V_2}{V_1}$$

$P_1 V_1$ can be replaced by $P_2 V_2$ or mRT_0 . Also, V_2/V_1 can be replaced by P_1/P_2 for this case since $P_1 V_1 = P_2 V_2$.

Substituting the numerical values

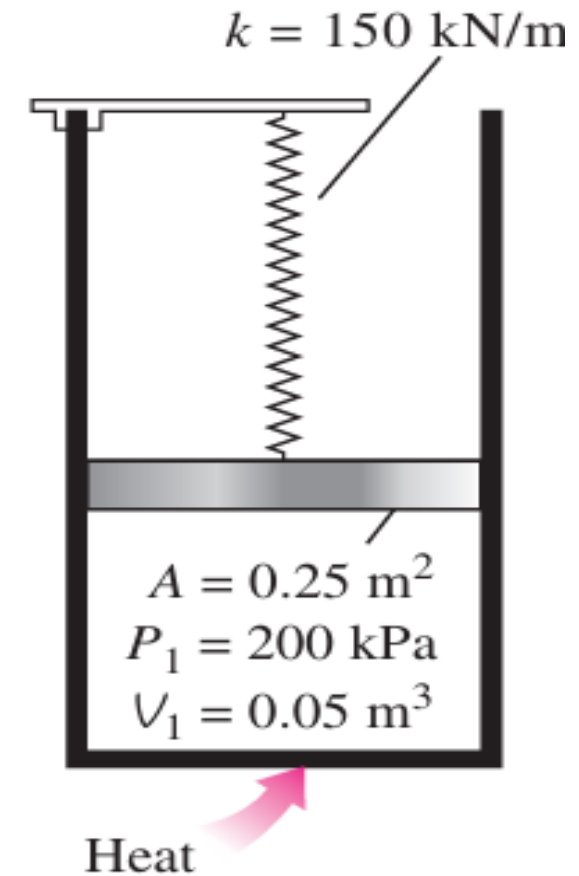
$$\begin{aligned} W_b &= (100 \text{ kPa})(0.4 \text{ m}^3) \left(\ln \frac{0.1}{0.4} \right) \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right) \\ &= -55.5 \text{ kJ} \end{aligned}$$

Discussion The negative sign indicates that this work is done on the system (a work input), which is always the case for compression processes.



Example:3.3

A piston–cylinder device contains 0.05 m^3 of a gas initially at 200 kPa . At this state, a linear spring that has a spring constant of 150 kN/m is touching the piston but exerting no force on it. Now heat is transferred to the gas, causing the piston to rise and to compress the spring until the volume inside the cylinder doubles. If the cross-sectional area of the piston is 0.25 m^2 , determine (a) the final pressure inside the cylinder, (b) the total work done by the gas, and (c) the fraction of this work done against the spring to compress it.



Solution A gas in a piston–cylinder device equipped with a linear spring expands as a result of heating. The final gas pressure, the total work done, and the fraction of the work done to compress the spring are to be determined.

Assumptions **1** The expansion process is quasi-equilibrium. **2** The spring is linear in the range of interest.

Analysis A sketch of the system and the P - V diagram of the process are shown in Fig. 2–10.

(a) The enclosed volume at the final state is

$$V_2 = 2V_1 = (2)(0.05 \text{ m}^3) = 0.1 \text{ m}^3$$

Then the displacement of the piston (and of the spring) becomes

$$x = \frac{\Delta V}{A} = \frac{(0.1 - 0.05) \text{ m}^3}{0.25 \text{ m}^2} = 0.2 \text{ m}$$

The force applied by the linear spring at the final state is

$$F = kx = (150 \text{ kN/m})(0.2 \text{ m}) = 30 \text{ kN}$$

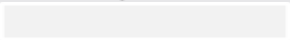
The additional pressure applied by the spring on the gas at this state is

$$P = \frac{F}{A} = \frac{30 \text{ kN}}{0.25 \text{ m}^2} = 120 \text{ kPa}$$

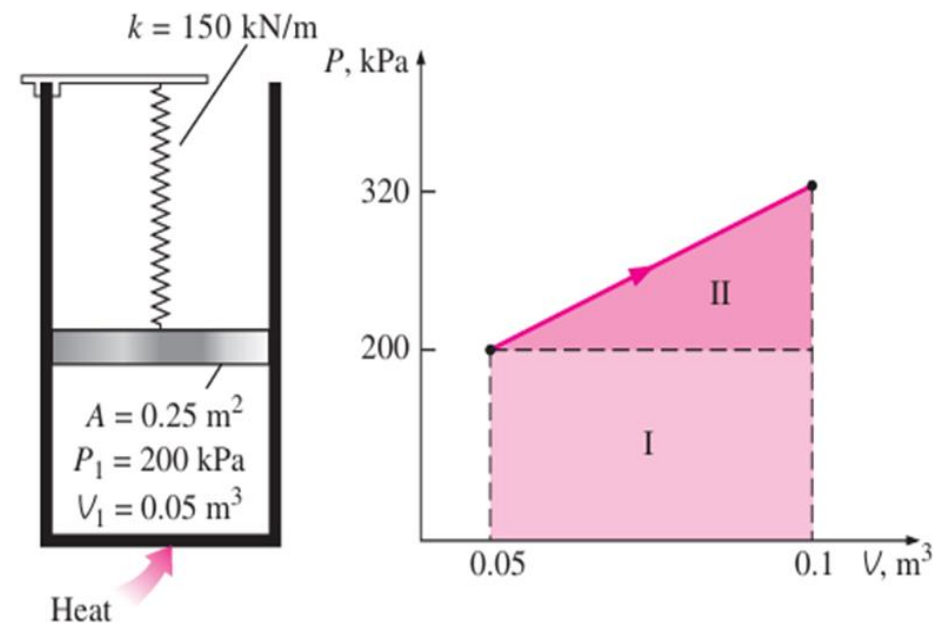
Without the spring, the pressure of the gas would remain constant at 200 kPa while the piston is rising. But under the effect of the spring, the pressure rises linearly from 200 kPa to

$$200 + 120 = \mathbf{320 \text{ kPa}}$$

at the final state.

(b) An easy way of finding the work done is to plot the process on a P - V diagram and find the area under the process curve.  the area under the process curve (a trapezoid) is determined to be

$$W = \text{area} = \frac{(200 + 320) \text{ kPa}}{2} [(0.1 - 0.05) \text{ m}^3] \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right) = \mathbf{13 \text{ kJ}}$$



Note that the work is done by the system.

(c) The work represented by the rectangular area (region I) is done against the piston and the atmosphere, and the work represented by the triangular area (region II) is done against the spring. Thus,

$$W_{\text{spring}} = \frac{1}{2}[(320 - 200) \text{ kPa}](0.05 \text{ m}^3) \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right) = \mathbf{3 \text{ kJ}}$$

Discussion This result could also be obtained from

$$W_{\text{spring}} = \frac{1}{2}k(x_2^2 - x_1^2) = \frac{1}{2}(150 \text{ kN/m})[(0.2 \text{ m})^2 - 0^2] \left(\frac{1 \text{ kJ}}{1 \text{ kN} \cdot \text{m}} \right) = 3 \text{ kJ}$$

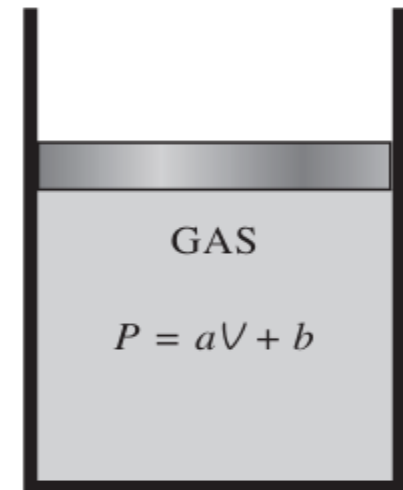
Example:3.4

What is polytropic index in the expansion of piston from

$P_1 = 120 \text{ Kpa}$ and $V_1 = 1.0 \text{ m}^3$ to $P_2 = 15 \text{ Kpa}$ and $V_2 = 4.0 \text{ m}^3$

Exercise:1

A gas is compressed from an initial volume of 0.42m^3 to a final volume of 0.12m^3 . During the quasi-equilibrium process, the pressure changes with volume according to the relation $P = aV + b$, where $a = -1200 \text{ kPa}/\text{m}^3$ and $b = 600 \text{ kPa}$. Calculate the work done during this process (a) by plotting the process on a P-V diagram and finding the area under the process curve and (b) by performing the necessary integrations.



Exercise:2

During some actual expansion and compression processes in piston–cylinder devices, the gases have been observed to satisfy the relationship $PV^n = C$, where n and C are constants. Calculate the work done when a gas expands from 150 kPa and 0.03 m^3 to a final volume of 0.2 m^3 for the case of $n = 1.3$.